IFN619 :: UA2 - Assignment 2 - Applied Data Analysis (60%) - PART A

IMPORTANT: Refer to the instructions in Canvas (`UA2 - Assignment 2`) *BEFORE* working on this assignment. Ensure that you are familiar with the marking rubric and understand how the grade for this assignment will be awarded according to the criteria.

1. Complete and run the code cell below to display your name and student number
2. Complete an appropriate number of full data analytics cylces which include structured and semi-structured data analysis (either separately or intermixed as relevant to your questions)

3. Ensure that you use ONLY the techniques and libraries/packages that have been used in class. Unless you are consolidating your understanding of foundational techniques, you must use techniques introduced in module C.
4. Ensure that you have addressed all of the criteria in the assignment rubric
5. Ensure that your notebook runs completey without errors
6. Clear all cells and re-run your entire notebook so that cells are sequentially numbered without any errors. **Not doing this step risks having your work marked as incomplete!**
7. Submit the clean notebook to Canvas **before** 11:55pm on the due date

ESSENTIAL REQUIREMENTS: Your task as a data analyst is to:

- Ensure that you use the techniques and libraries/packages that have been used in class
- Identify high quality questions that when answered may be helpful in addressing the provided scenario
- Clean and filter the data as appropriate
- Analyse the data in a way that answers important questions and ultimately addresses the concern in the scenario
- Visualise your results in a meaningful way that is helpful in making visible key findings
- Provide a detailed summary of the insights found and how they address the original questions and scenario
- Ensure that your analysis provides an effective narrative that directly addresses the scenario and answers the relevant questions
- Ensure that you have taken account of important ethical considerations directly (and specifically) relevant to your analysis

AUTHENTICITY AND INTEGRITY: You will be marked on (a) *HOW* you undertake the task todgether; with (b) detail of *WHY* you made various decisions involved in the tasks; and (c) acknowledgement of **WHERE** you used material that is not directly yours. Therefore, **you must document your thinking and approach throughout the notebook using the Markdown cells, and give credit to other resources as appropriate.** You are encouraged to use the `Exemplars` PDF to help write your code. You may use online resources including `GenAI tools` and `stackoverflow` to help you write your code, however you must acknowledge that

you are using these resources in the markdown cells explaining your analysis. Note that you do not need to use formal referencing for this.

NOTE FOR CONSOLIDATION OPTION STUDENTS ONLY

If you choose to consolidate your foundational techniques (understanding you cannot achieve full marks for this assessment), you will demonstrate effective use of the techniques from module A only and will not use additional techniques from module C. You are still required to include the ethical considerations.

Ensure that you set `consolidation_option=True` *in the cell below to ensure that you are marked correctly for this assignment.*

```
# Complete the following cell with your details and run to produce
your personalised header for this assignment

from IPython.display import HTML

first_name = "Srinivasa Teja Bharadwaj"
last_name = "Penumudi"
student_number = [redacted]
consolidation_option = False # Change to true if you are choosing
consolidation option

personal_header = f"<h1>{first_name} {last_name}
({student_number})</h1>"
if consolidation_option:
    personal_header+="<h2>CONSOLIDATION SELECTED</h2>"
HTML(personal_header)
```

Scenario for Assignment 1 and Assignment 2 (Part A)

You are working as a Data Scientist assisting the Australian Sports Commission (ASC) in the lead-up to the Brisbane 2032 Olympic Games.

A recommendation needing investigation is for the ASC to build fan support for athletes over the next 6 years through strategically connecting with school age children and youth.

Your task is to investigage how relevant data might inform this strategy, particularly by taking advantage of the following suggestions:

1. Some school age children and yout (now) are likely to be Olympians in some sports offered in 2032. Tapping this peer connection could be a valuable part of the ASC strategy.

2. Some sports are highly popular with school age children and youth actively participating in them. There is potential to build fan support between athletes competing in these sports and kids participating in them.
3. Early adolescents will begin to use social media more in the next few years, and then be adults when the games occur. Connecting with this group may enable a steady build of wider social support which could boost the ASC strategy.

Applied Data Analysis Task

Extend and refine your analysis from assignment 1, by using additional techniques (from module C), and expanding your selection of data. Select appropriate additional data (from the AusPlay data) to develop your analysis further.

Ensure that you present a clear narrative connecting your analysis to the scenario and important questions derived from the suggestions in the scenario.

You should also ensure that you include important and relevant ethical considerations throughout your analysis.

DATA:

Select relevant structured data from: AusPlay results

Select relevant semi/unstructured data from: Guardian API

<<<< INSERT CELLS BELOW >>>>

1. INTRODUCTION

The Brisbane 2032 Olympic Games present a major opportunity for the Australian Sports Commission (ASC) to strengthen youth engagement and long-term sports participation across Australia. Understanding which sports attract younger demographics, and how sports are represented within media visibility patterns, may help identify opportunities for future fan engagement strategies.

This analysis combines structured participation data from AusPlay with semi/unstructured media data collected through the Guardian API. By analysing both participation patterns and public media discussions, the notebook aims to identify sports that may have strong potential for future youth engagement leading into Brisbane 2032.

The analysis applies advanced techniques introduced in the second half of the unit, including clustering, semi-structured analytics, narrative analytics, and ethical considerations. These techniques are used to develop meaningful insights that may support strategic sports engagement planning and understanding of public sports discourse

RESEARCH QUESTION

How can the Australian Sports Commission use youth participation patterns and media visibility patterns to identify priority sports and engagement opportunities ahead of the Brisbane 2032 Olympic Games?

Why This Research Question Was Chosen

This research question was chosen because the Brisbane 2032 Olympic Games provide an important opportunity for the Australian Sports Commission (ASC) to strengthen youth engagement and long-term participation in sport across Australia. Understanding which sports attract younger demographics, and how these sports are represented in public media, may help support future participation and engagement strategies.

The analysis combines structured AusPlay participation data with semi-structured Guardian API media data. AusPlay data helps identify youth participation trends and demographic engagement patterns, while Guardian media data provides insight into how sports are represented within public sporting narratives.

By integrating both datasets, the investigation explores possible relationships between youth participation behaviour and media visibility across different sports. This may help ASC better identify sports with strong future engagement potential leading into Brisbane 2032, as well as sports that may require additional promotion or development support.

This research question also extends the work completed in UA1 by introducing more advanced analytical techniques, including clustering, semi-structured analytics, TF-IDF analysis, and ethical considerations. These techniques support a deeper and more strategic investigation compared to the earlier assignment, which focused mainly on descriptive analysis and visualisation.

Analysis plan (QDAVI) Framework

This notebook follows a QDAVI-style analytical framework consisting of Question, Data, Analysis, Visualisation, Insights, and Ethical Considerations. This structure was chosen to create a clear analytical workflow and ensure that each stage of the investigation contributes meaningfully towards answering the research question.

Question

The analysis focuses on understanding how youth participation patterns and media visibility patterns can help identify sports with strong engagement potential leading into the Brisbane 2032 Olympic Games.

Data

The investigation combines structured participation data from AusPlay with semi/unstructured media visibility patterns data collected through the Guardian API. The combination of these datasets allows both behavioural participation analysis and public discussion analysis to be explored together.

Analysis

The analysis will apply techniques introduced in the second half of the unit, including clustering, semi-structured analytics, comparative analysis, and narrative analytics. These techniques were selected to move beyond descriptive statistics and generate deeper strategic insights.

Visualisation

Interactive Plotly visualisations will be used to identify participation trends, demographic patterns, and relationships between participation behaviour and media attention across different sports.

Insights

The analysis aims to identify sports with strong youth engagement potential, possible gaps between participation and media attention, and opportunities for future sports promotion and community engagement strategies for the ASC and Brisbane 2032 stakeholders.

Ethical Considerations

Ethical considerations will be integrated throughout the notebook, including discussion of media bias, limitations of public discourse data, representation issues, and interpretation risks associated with combining structured and semi/unstructured datasets.

2. Data Selection and Preparation

The first stage of the analysis focuses on preparing and understanding the structured participation datasets collected from AusPlay. These datasets provide national participation statistics across different sports, demographic groups, and participation behaviours.

The structured datasets were selected because they support the investigation of youth participation trends, sport popularity, and engagement patterns relevant to the Brisbane 2032 Olympic Games. Understanding these patterns is important for identifying sports with strong future engagement potential and for supporting strategic planning by the Australian Sports Commission (ASC) and the Queensland Government.

At this stage, the datasets will be imported and explored to understand their structure, available variables, and relevance to the research question before deeper analysis techniques are applied later in the notebook.

```
# Data handling
import pandas as pd
import numpy as np

# Text processing
import re

# Visualisation
```

```
import plotly.express as px
import plotly.graph_objects as go

# Machine learning / clustering
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler

# Warning control
import warnings
warnings.filterwarnings('ignore')
```

3. Structured AusPlay Participation Analysis

The structured analysis focuses specifically on Olympic-relevant and high-engagement sports associated with youth participation. Rather than analysing every activity available within the AusPlay dataset, the investigation focuses on selected sports that are strategically relevant to the Brisbane 2032 Olympic Games and future youth engagement initiatives.

The selected sports include activities such as athletics, basketball, cycling, swimming, football, tennis, volleyball, netball, martial arts, cricket, and other Olympic-related sports with strong youth participation relevance. Focusing on these sports helps reduce unnecessary noise within the dataset and allows the analysis to concentrate on activities that are more meaningful for long-term Olympic audience development and participation strategy.

The structured dataset also includes participation rates across multiple youth age groups and gender categories. These variables are important because they support deeper analytical techniques introduced in the second half of the unit, including clustering and comparative participation analysis.

```
# ============================================================
# LOAD ONLY THE RELEVANT CHILDREN PARTICIPATION SECTION
# ============================================================

ausplay_df = pd.read_excel(
    "C4S-AusPlay-By-Sport-Data-Tables-13-November-2025.xlsx",

    # Table 2 = Participation by activity (children)
    sheet_name="2",

    # Skip heading rows before actual data starts
    skiprows=158,

    # Read only the participation rows
    nrows=142,

    # Remove default Excel headers
    header=None,

    # Assign custom column names
```

```
    names=[
        "sport",
        "total",
        "C",
        "males",
        "females",
        "G",
        "0-4",
        "5-8",
        "9-11",
        "12-14"
    ]
)

# Display dataset preview
ausplay_df.head()

# Remove empty formatting columns
ausplay_df = ausplay_df.drop(columns=["C", "G"])

# Rename age columns for cleaner use
ausplay_df = ausplay_df.rename(columns={
    "0-4": "age_0_4",
    "5-8": "age_5_8",
    "9-11": "age_9_11",
    "12-14": "age_12_14"
})

# Preview cleaned data
ausplay_df.head()
```

This table contains estimated participation counts for children by activity. The unused columns C and G are removed because they are formatting gaps from the original AusPlay workbook and do not contain analytical values. The age-group columns are renamed to make them easier to use in later filtering, visualisation, and clustering steps.

```
# ============================================================
# FILTER OLYMPIC-RELEVANT SPORTS
# ============================================================

# Selected Olympic-relevant sports

olympic_sports = [
    "Athletics, track and field",
    "Australian football",
    "Basketball",
    "Cricket",
    "Gymnastics",
    "Hockey",
    "Martial arts",
```

```
    "Netball",
    "Swimming",
    "Table tennis",
    "Tennis",
    "Touch football",
    "Volleyball (indoor and outdoor)",
    "Badminton"
]

# Filter dataset

olympic_df = ausplay_df[
    ausplay_df["sport"].isin(olympic_sports)
]

# Reset index

olympic_df = olympic_df.reset_index(drop=True)

# Display filtered dataset

print("Olympic-Relevant Dataset Shape:")
print(olympic_df.shape)

display(olympic_df)
```

The dataset is filtered to include selected Olympic-relevant and high youth-engagement sports associated with Brisbane 2032. This focused approach helps reduce unnecessary noise within the dataset and allows the analysis to concentrate on sports that are strategically meaningful for future Olympic audience development and participation planning.

The selected sports represent activities with strong youth participation relevance, public recognition, and long-term Olympic engagement potential. Focusing on these sports also improves the effectiveness of later analytical techniques such as clustering and comparative participation analysis.

4. Youth Engagement Score and Visualisation

While total participation provides a general indication of sport popularity, it does not fully reflect long-term youth engagement potential relevant to Brisbane 2032. Since this investigation focuses on future Olympic participation and audience development, greater emphasis is placed on school-age and early adolescent participation.

To better represent sustained youth involvement, a combined youth engagement score is created using the 5–8, 9–11, and 12–14 age groups. These categories are important because they reflect key developmental stages associated with long-term sport participation and future Olympic engagement.

Creating this aggregated variable also extends the analysis beyond basic descriptive reporting by introducing a new analytical feature that supports ranking, visualisation, and clustering analysis. This helps identify which Olympic-relevant sports demonstrate the strongest youth engagement potential leading into Brisbane 2032.

```
# Select Olympic-relevant and high youth participation sports

olympic_sports = [
    "Swimming",
    "Basketball",
    "Athletics, track and field",
    "Tennis",
    "Gymnastics",
    "Badminton",
    "Table tennis",
    "Hockey",
    "Archery",
    "Cricket",
    "Netball",
    "Martial arts",
    "Volleyball (indoor and outdoor)",
    "Touch football"
]

# Filter selected sports
olympic_df = ausplay_df[
    ausplay_df["sport"].isin(olympic_sports)
]

# Reset index
olympic_df = olympic_df.reset_index(drop=True)

# Preview dataset
display(olympic_df.head())

# ========================================
# CREATE YOUTH ENGAGEMENT SCORE
# ========================================

# Create combined engagement score
olympic_df["youth_engagement_score"] = (
    olympic_df[["age_5_8", "age_9_11", "age_12_14"]]
    .sum(axis=1)
)

# Rank sports based on engagement score
engagement_ranking = olympic_df.sort_values(
    by="youth_engagement_score",
    ascending=False
)
```

```
# Display rankings
display(
    engagement_ranking[
        [
            "sport",
            "youth_engagement_score",
            "males",
            "females"
        ]
    ].head(10)
)

# Round values for cleaner presentation
engagement_ranking = engagement_ranking.round(3)

# Display rounded results
display(
    engagement_ranking[
        [
            "sport",
            "youth_engagement_score",
            "males",
            "females"
        ]
    ].head(10)
)
```

Ranking and Presentation of Youth Engagement Results

After constructing the Youth Engagement Score, the dataset was ranked in descending order to identify sports with the strongest levels of sustained youth participation. Ranking the sports using the combined engagement metric allows clearer comparison between Olympic-relevant activities and highlights sports with higher participation among children aged 5–14 years.

To improve readability, numerical values were rounded to three decimal places. This does not affect the analysis but helps present the participation metrics more clearly within tables and visualisations.

The ranked results provide a clearer overview of which sports demonstrate stronger youth engagement potential and support further analysis through visualisation, demographic comparison, and clustering techniques.

Visualising Youth Engagement Across Olympic-Relevant Sports

To better interpret participation trends across Olympic-relevant sports, an interactive visualisation of Youth Engagement Scores was developed using Plotly.

Interactive visualisations improve analytical communication by allowing clearer comparison between sports while presenting participation differences in a more intuitive and visually

interpretable format. A horizontal bar chart was selected because it improves readability when comparing multiple sports with long category names.

The visualisation supports the identification of:

- sports with exceptionally high youth engagement,
- participation intensity differences across activities,
- and potential strategic participation patterns associated with Brisbane 2032.

This analytical step transforms numerical rankings into interpretable visual insights that support evidence-based discussion and comparative participation analysis.

```
# ============================================
# INTERACTIVE YOUTH ENGAGEMENT VISUALISATION
# ============================================

import plotly.express as px

# Sort data for visualisation
plot_data = engagement_ranking.sort_values(
    by="youth_engagement_score",
    ascending=True
)

# Create interactive horizontal bar chart
fig = px.bar(
    plot_data,
    x="youth_engagement_score",
    y="sport",
    orientation="h",
    color="youth_engagement_score",
    title="Youth Engagement Across Olympic-Relevant Sports",
    labels={
        "youth_engagement_score": "Youth Engagement Score",
        "sport": "Sport"
    },
    height=700
)

# Improve layout
fig.update_layout(
    template="plotly_white",
    title_x=0.5
)

# Display chart
fig.show()
```

Interpretation of Youth Engagement Visualisation

The visualisation highlights clear differences in youth participation across Olympic-relevant sports. Swimming demonstrates the strongest overall youth engagement, significantly outperforming other selected sports. This suggests that swimming maintains consistently high participation across multiple youth age groups and remains one of the strongest long-term sporting pathways in Australia.

Basketball also demonstrates strong youth engagement, indicating growing popularity and accessibility among younger demographics. Sports such as Netball, Gymnastics, Tennis, and Athletics also show sustained participation patterns, likely supported by school programs, junior development pathways, and community-level accessibility.

In contrast, sports such as Archery, Table Tennis, and Badminton display comparatively lower engagement scores, suggesting smaller youth participation bases relative to higher-engagement sports.

Overall, the visualisation demonstrates that youth participation is not evenly distributed across Olympic-relevant activities. These findings provide useful insight for ASC planning by identifying sports with stronger grassroots participation and long-term engagement potential leading into Brisbane 2032.

5. Gender and Age Group Participation Analysis

To further explore participation behaviour across Olympic-relevant sports, a comparative gender analysis was conducted using male and female participation rates.

Analysing participation by gender provides deeper insight into how sporting engagement differs across demographic groups and helps identify sports with balanced participation, male-dominated engagement, or stronger female participation patterns.

This analysis is important because participation inequalities and demographic participation preferences may influence:

- long-term sporting sustainability,
- accessibility,
- youth development pathways,
- and future participation planning associated with Brisbane 2032.

By comparing male and female participation levels across sports, the analysis aims to identify broader engagement trends and evaluate how participation behaviour differs between demographic categories.

```
# ========================================
# GENDER PARTICIPATION COMPARISON
# ========================================

import plotly.graph_objects as go
```

```
# Create figure
fig = go.Figure()

# Male participation
fig.add_trace(
    go.Bar(
        x=olympic_df["sport"],
        y=olympic_df["males"],
        name="Male Participation"
    )
)

# Female participation
fig.add_trace(
    go.Bar(
        x=olympic_df["sport"],
        y=olympic_df["females"],
        name="Female Participation"
    )
)

# Update layout
fig.update_layout(
    title="Male vs Female Participation Across Olympic-Relevant
Sports",
    xaxis_title="Sport",
    yaxis_title="Participation Rate",
    barmode="group",
    template="plotly_white",
    height=700
)

# Show chart
fig.show()
```

Analysis of Gender Participation Patterns

The visualisation highlights clear differences in male and female participation across Olympic-relevant sports. Swimming demonstrates the highest overall participation for both genders, suggesting strong accessibility and broad youth engagement across demographic groups.

Basketball and cricket show stronger male participation rates, while gymnastics and netball display substantially higher female participation. These patterns may reflect differences in sporting culture, grassroots development pathways, and long-standing participation preferences within Australian youth sport.

Sports such as athletics, tennis, badminton, and touch football demonstrate comparatively balanced participation levels, indicating broader accessibility across genders.

Overall, the findings suggest that youth sporting engagement is not evenly distributed between demographic groups. These participation patterns provide useful insight for ASC planning by identifying sports with stronger inclusiveness, participation balance, and long-term youth engagement potential leading into Brisbane 2032.

Age Group Participation Analysis

To further understand youth engagement patterns, participation across different age groups was analysed. This step helps identify whether selected Olympic-relevant sports attract younger children, older children, or maintain participation consistently across multiple developmental stages.

Age-group analysis is important because youth engagement is not only about total participation. A sport may have high participation among younger children but weaker retention among older age groups, or it may show stronger engagement as children move closer to adolescence.

By comparing the 0–4, 5–8, 9–11, and 12–14 age groups, this analysis helps identify sports with strong early exposure, sustained participation, and possible long-term engagement potential leading into Brisbane 2032.

```
# ========================================
# AGE GROUP PARTICIPATION ANALYSIS
# ========================================

age_group_data = olympic_df.melt(
    id_vars="sport",
    value_vars=["age_0_4", "age_5_8", "age_9_11", "age_12_14"],
    var_name="age_group",
    value_name="participation_rate"
)

fig = px.bar(
    age_group_data,
    x="sport",
    y="participation_rate",
    color="age_group",
    barmode="group",
    title="Youth Participation by Age Group Across Olympic-Relevant
Sports",
    labels={
        "sport": "Sport",
        "participation_rate": "Participation Rate",
        "age_group": "Age Group"
    },
    height=700
)

fig.update_layout(
    template="plotly_white",
```

```
    xaxis_tickangle=-35,
    title_x=0.5
)

fig.show()
```

Interpretation of Age Group Participation Trends

The visualisation highlights clear participation differences across youth age groups within Olympic-relevant sports. Swimming demonstrates the strongest overall participation across all age categories, particularly among ages 5–8 and 9–11, suggesting strong accessibility and long-term engagement potential.

Basketball, gymnastics, and netball also show strong participation patterns, although the dominant age groups vary. Gymnastics appears more popular among younger children, while basketball participation increases more noticeably within older youth categories.

Sports such as archery, table tennis, and hockey display comparatively lower participation across all age groups, which may reflect lower accessibility or reduced grassroots exposure compared to more mainstream sports.

Overall, the findings suggest that participation behaviour changes across developmental stages and varies significantly between sports. These patterns provide useful insight for ASC planning by identifying sports with stronger long-term participation pathways and youth engagement potential leading into Brisbane 2032.

6. Clustering and Strategic Segmentation

While earlier analyses focused on comparing individual sports, clustering analysis is used to identify broader participation patterns within the dataset. Analysing sports individually can make it difficult to recognise deeper behavioural similarities, whereas clustering helps group sports with similar participation characteristics across demographic categories.

K-Means clustering is applied because it is an effective unsupervised machine learning technique for identifying natural groupings within numerical data. The analysis uses variables including male participation, female participation, and youth age-group engagement to determine whether certain Olympic-relevant sports share similar participation structures.

From an analytical perspective, clustering helps simplify complex participation data into clearer engagement categories while also supporting pattern discovery beyond standard descriptive analysis. Sports grouped within high-engagement clusters may represent stronger grassroots participation foundations, while lower-engagement clusters may indicate niche or emerging sports with different participation characteristics.

Overall, clustering extends the investigation beyond simple participation rankings and provides more strategic insight into youth engagement patterns relevant to ASC planning and Brisbane 2032.

```
# ========================================
# PREPARE DATA FOR CLUSTERING
# ========================================

from sklearn.preprocessing import StandardScaler

# Select numerical participation features
cluster_features = olympic_df[
    [
        "males",
        "females",
        "age_0_4",
        "age_5_8",
        "age_9_11",
        "age_12_14"
    ]
]

# Standardize values
scaler = StandardScaler()

scaled_features = scaler.fit_transform(cluster_features)

# Preview scaled data shape
scaled_features.shape

# ========================================
# APPLY K-MEANS CLUSTERING
# ========================================

from sklearn.cluster import KMeans

# Create KMeans model
kmeans = KMeans(
    n_clusters=3,
    random_state=42
)

# Fit model and predict clusters
olympic_df["cluster"] = kmeans.fit_predict(scaled_features)

# Preview cluster assignments
display(
    olympic_df[
        [
            "sport",
            "cluster"
        ]
    ].sort_values("cluster")
)
```

Interpretation of Cluster Groupings

The clustering results reveal clear participation segments across Olympic-relevant sports based on youth engagement behaviour. The K-Means algorithm identified three distinct clusters using demographic participation variables across gender and age-group categories.

Swimming forms an independent cluster, indicating exceptionally strong and consistent participation across multiple youth demographics. This suggests that swimming maintains a uniquely high level of grassroots engagement compared to other sports in the dataset.

Basketball and netball form a second high-engagement cluster, reflecting strong youth participation and broad accessibility. These sports appear to maintain more sustained participation patterns than most remaining activities.

The remaining sports form a larger cluster representing moderate or lower participation structures. While these sports remain Olympic-relevant, their engagement patterns appear more specialised or comparatively smaller across demographic groups.

Overall, the clustering process simplifies complex participation data into clearer behavioural segments and provides more strategic insight than analysing sports individually. These findings help identify sports with stronger long-term participation potential relevant to ASC planning and Brisbane 2032.

```
# ========================================
# CORRELATION HEATMAP
# ========================================

# Select variables for correlation analysis
correlation_data = olympic_df[
    [
        "males",
        "females",
        "age_0_4",
        "age_5_8",
        "age_9_11",
        "age_12_14",
        "youth_engagement_score"
    ]
]

# Calculate correlation matrix
corr_matrix = correlation_data.corr()

# Create Plotly heatmap
fig = px.imshow(
    corr_matrix,
    text_auto=True,
    color_continuous_scale="RdBu_r",
    title="Correlation Heatmap of Youth Participation Variables",
    labels=dict(color="Correlation")
```

```
)

fig.update_layout(
    template="plotly_white",
    title_x=0.5,
    height=700
)

fig.show()
```

Interpretation of Correlation Heatmap

The correlation heatmap reveals strong positive relationships across most participation variables within the dataset. Sports with high participation in one demographic category generally also demonstrate strong participation across other age groups and genders.

One of the strongest relationships appears between the youth engagement score and the 9–11 age group, suggesting that this developmental stage contributes significantly to overall youth participation patterns. Strong positive correlations are also visible between the 5–8 and 9–11 age groups, indicating consistent participation continuity as children grow older.

Male and female participation variables also show strong positive relationships with overall engagement scores, suggesting that highly engaged sports tend to attract participation from both genders rather than a single dominant demographic group.

Overall, the findings confirm that youth participation variables are highly interconnected across the dataset. These results strengthen the earlier clustering analysis and provide useful insight for ASC planning by identifying sports with stronger long-term participation sustainability and broader grassroots engagement potential leading into Brisbane 2032.

```
# ========================================
# CLUSTER PROFILE ANALYSIS
# ========================================

# Calculate average values for each cluster
cluster_profile = olympic_df.groupby("cluster")[
    [
        "males",
        "females",
        "age_0_4",
        "age_5_8",
        "age_9_11",
        "age_12_14",
        "youth_engagement_score"
    ]
].mean()

# Round values for readability
cluster_profile = cluster_profile.round(3)
```

```python
# Display cluster profiles
display(cluster_profile)

# ==========================================
# VISUALISE CLUSTER PROFILES
# ==========================================

profile_data = cluster_profile.reset_index()

fig = px.bar(
    profile_data,
    x="cluster",
    y="youth_engagement_score",
    color="cluster",
    title="Average Youth Engagement Score by Cluster",
    labels={
        "cluster": "Cluster Group",
        "youth_engagement_score": "Average Youth Engagement Score"
    },
    height=600
)

fig.update_layout(
    template="plotly_white",
    title_x=0.5
)

fig.show()

# ==========================================
# STRATEGIC SPORT SEGMENTATION
# ==========================================

# Create engagement categories
conditions = [
    olympic_df["youth_engagement_score"] >= 0.20,
    olympic_df["youth_engagement_score"] >= 0.08,
    olympic_df["youth_engagement_score"] < 0.08
]

categories = [
    "High Engagement",
    "Medium Engagement",
    "Emerging Participation"
]

# Assign categories
olympic_df["engagement_category"] = np.select(
    conditions,
    categories,
```

```
    default="Emerging Participation"
)

# Preview segmentation
display(
    olympic_df[
        [
            "sport",
            "youth_engagement_score",
            "engagement_category"
        ]
    ].sort_values(
        by="youth_engagement_score",
        ascending=False
    )
)

# ========================================
# VISUALISE STRATEGIC SEGMENTS
# ========================================

fig = px.bar(
    olympic_df.sort_values(
        by="youth_engagement_score",
        ascending=False
    ),
    x="sport",
    y="youth_engagement_score",
    color="engagement_category",
    title="Strategic Segmentation of Olympic-Relevant Sports",
    labels={
        "sport": "Sport",
        "youth_engagement_score": "Youth Engagement Score",
        "engagement_category": "Engagement Category"
    },
    height=700
)

fig.update_layout(
    template="plotly_white",
    title_x=0.5,
    xaxis_tickangle=-35
)

fig.show()
```

7. Correlation Analysis

Following the clustering and engagement analyses, a strategic segmentation approach was applied to classify Olympic-relevant sports based on their overall youth engagement performance. Instead of treating all sports equally, this step groups activities into clearer participation categories using the calculated youth engagement scores.

The purpose of this segmentation is to simplify complex participation patterns into more interpretable strategic groups. Sports with consistently strong engagement across multiple youth age groups are classified as high-engagement sports, while activities with lower or developing participation patterns are grouped separately.

From an analytical perspective, segmentation helps transform raw participation data into clearer behavioural categories and supports stronger comparison between sports. This provides useful insight into which sports currently demonstrate stronger grassroots participation foundations and which activities may require additional promotion or youth development support.

Overall, the segmentation analysis provides a more strategic understanding of youth sporting engagement patterns and supports ASC planning related to long-term participation and Brisbane 2032 audience development.

Ethical Considerations of Clustering Analysis

While clustering analysis helps identify participation patterns across Olympic-relevant sports, the results should be interpreted carefully. K-Means clustering groups sports based only on numerical similarity and does not consider broader social or environmental factors such as accessibility, infrastructure, cultural preferences, or economic barriers that may influence youth participation.

There is also a risk of oversimplifying participation behaviour by grouping sports into broad engagement categories. Lower participation rates do not necessarily indicate lower importance, as some sports may face structural or development limitations rather than reduced interest.

In addition, clustering outcomes can vary depending on the selected variables, scaling methods, and number of clusters used in the model. Therefore, the results should be viewed as exploratory analytical insights rather than definitive classifications.

Overall, the clustering analysis should support inclusive and responsible sporting development while recognising the broader context surrounding youth participation and Brisbane 2032 engagement planning.

Semi Structured Analysis

Research Question

How can the Australian Sports Commission use youth participation patterns and media visibility patterns to identify priority sports and engagement opportunities ahead of the Brisbane 2032 Olympic Games?

Why this question matters

While the structured participation analysis identified sports with strong youth engagement patterns, participation statistics alone do not fully explain how Olympic-related sports are represented within broader public sporting narratives. To extend the investigation, semi-structured media analysis is introduced using sports-related article headlines collected through The Guardian API.

The semi-structured component of the investigation supports the following research objective:

> How can the Australian Sports Commission (ASC) use youth participation patterns and media visibility patterns to identify priority sports and engagement opportunities ahead of the Brisbane 2032 Olympic Games?

Unlike structured participation data, media headlines provide insight into the visibility, prominence, and recurring narratives surrounding Olympic sports, athletes, and major sporting events. Analysing these media patterns may help identify which sports currently receive stronger public attention and how media exposure may influence long-term sporting interest and youth engagement.

From a stakeholder perspective, this analysis may assist the ASC and Brisbane 2032 stakeholders in understanding how participation behaviour and media visibility interact within the broader sporting ecosystem. Combining structured participation analytics with exploratory media analysis therefore provides a more holistic evidence-based framework for supporting youth participation planning, sport promotion strategies, and long-term Olympic audience development.

8. Guardian API Media Analytics

To extend the analysis beyond structured participation statistics, external media data from The Guardian Open Platform API is integrated into the investigation. While the AusPlay dataset provides quantitative insight into youth participation behaviour, it does not explain broader public visibility, media attention, or narrative trends surrounding Olympic-relevant sports.

The inclusion of Guardian API data introduces a semi-structured analytics component into the project by incorporating real-world news and media content related to sport participation, Olympic discussion, and public sporting engagement.

From an analytical perspective, combining participation data with media coverage allows deeper exploration into whether sports receiving stronger public and media attention also

demonstrate stronger youth participation patterns. This supports a more strategic interpretation of Olympic audience development and broader sporting engagement trends relevant to Brisbane 2032.

The Guardian API returns data in JSON format, which represents semi-structured data containing attributes such as article headlines, publication dates, section names, and article URLs. These attributes can be transformed into a structured analytical dataset suitable for comparative and exploratory analysis.

```
# ==================================================
# IMPORT REQUIRED LIBRARIES
# ==================================================

import pandas as pd
import requests
import json
import re

import plotly.express as px
import plotly.graph_objects as go

from sklearn.feature_extraction.text import TfidfVectorizer

# Load Guardian API key safely

with open("Guardian_api_key.txt", "r") as file:
    api_key = file.read().strip()

print("Guardian API Key Loaded Successfully")

# ==================================================
# CREATE GUARDIAN SEARCH QUERY
# ==================================================

guardian_query = (
    '("Brisbane 2032" OR Olympics OR Olympic OR '
    '"youth sport" OR "young athletes") '
    'AND (swimming OR basketball OR tennis OR athletics '
    'OR gymnastics OR cricket OR netball)'
)

guardian_query

# ==================================================
# CONNECT TO GUARDIAN API
# ==================================================

url = "https://content.guardianapis.com/search"

params = {
```

```
    "api-key": api_key,
    "q": guardian_query,
    "section": "sport",
    "page-size": 100,
    "show-fields": "headline,trailText,bodyText",
    "from-date": "2023-01-01",
    "to-date": "2026-12-31",
    "order-by": "relevance"
}

# Send request to Guardian API
response = requests.get(url, params=params)

# Convert response into JSON format
guardian_data = response.json()

# Check API response status
guardian_data["response"]["status"]

# ==================================================
# CHECK TOTAL ARTICLES RETURNED
# ==================================================

guardian_data["response"]["total"]

# ==================================================
# EXTRACT ARTICLES INTO DATAFRAME
# ==================================================

articles = guardian_data["response"]["results"]

guardian_articles = []

for article in articles:

   guardian_articles.append({
    "headline": article["webTitle"],
    "section": article["sectionName"],
    "publication_date": article["webPublicationDate"],
    "url": article["webUrl"],
    "trail_text": article["fields"].get("trailText", ""),
    "body_text": article["fields"].get("bodyText", "")
    })

# Create dataframe
guardian_df = pd.DataFrame(guardian_articles)

# Preview dataset
guardian_df.head(10)
```

Media Data Cleaning and Standardisation

The Guardian API returns semi-structured textual data containing inconsistent formatting, punctuation, symbols, and mixed capitalization styles. Before conducting text analytics, the media headlines were cleaned and standardised to improve consistency and analytical reliability.

The preprocessing stage converted all headlines to lowercase and removed punctuation, special characters, and unnecessary formatting noise. This standardisation process helps ensure that words with identical meaning are treated consistently during analysis.

For example, terms such as “Olympic”, “olympic”, and “Olympic!” would otherwise be interpreted as separate tokens during text vectorisation. Cleaning the media data reduces analytical noise and improves the effectiveness of later techniques such as TF-IDF keyword extraction and clustering.

From a data analytics perspective, preprocessing is an essential step because textual media data is naturally unstructured and often contains inconsistencies that can negatively influence pattern detection and machine learning outcomes.

```
# ================================================
# CLEAN ARTICLE HEADLINES
# ================================================

# Convert headlines to lowercase
guardian_df["clean_headline"] = guardian_df["headline"].str.lower()

# Remove punctuation and special characters
guardian_df["clean_headline"] = guardian_df["clean_headline"].apply(
    lambda x: re.sub(r"[^a-zA-Z\s]", "", x)
)

# Preview cleaned headlines
guardian_df[
    ["headline", "clean_headline"]
].head(10)
```

```
import re

guardian_df["clean_headline"] = (
    guardian_df["headline"]
    .astype(str)
    .str.lower()
    .apply(lambda x: re.sub(r"[^a-zA-Z\s]", "", x))
)

guardian_df[["headline", "clean_headline"]].head()
```

9. Segregating Olympic-Related Media Articles

Before applying text analytics, the Guardian media dataset is filtered to separate Olympic-related articles from the broader sports article collection. This step helps reduce unrelated media noise and keeps the semi-structured analysis focused on sports, athletes, and Olympic themes relevant to Brisbane 2032.

This filtering supports the research question by ensuring that later techniques such as Count Vectorization, TF-IDF, and NMF topic modelling are applied to a more relevant subset of articles. For ASC stakeholders, this provides a clearer view of media visibility around Olympic-related sport engagement rather than general sports coverage.

```
# ================================================
# SEGREGATE OLYMPIC-RELATED ARTICLES
# ================================================

olympic_keywords = [
    "olympic", "olympics", "games", "athlete", "athletes",
    "athletics", "swimming", "basketball", "gymnastics",
    "tennis", "cricket", "netball", "hockey", "archery",
    "badminton", "table tennis", "volleyball", "brisbane"
]

keyword_pattern = "|".join(olympic_keywords)

olympic_articles_df = guardian_df[
    guardian_df["clean_headline"].str.contains(
        keyword_pattern,
        case=False,
        na=False
    )
].copy()

non_olympic_articles_df = guardian_df[
    ~guardian_df["clean_headline"].str.contains(
        keyword_pattern,
        case=False,
        na=False
    )
].copy()

print("Total Guardian articles:", len(guardian_df))
print("Olympic-related articles:", len(olympic_articles_df))
print("Other sports articles:", len(non_olympic_articles_df))

display(olympic_articles_df[["headline", "clean_headline"]].head(10))

# ================================================
# VISUALISE ARTICLE SEGMENTATION
# ================================================
```

```
article_segmentation = pd.DataFrame({
    "Article Category": ["Olympic-related articles", "Other sports
articles"],
    "Count": [
        len(olympic_articles_df),
        len(non_olympic_articles_df)
    ]
})

fig = px.bar(
    article_segmentation,
    x="Article Category",
    y="Count",
    text="Count",
    title="Guardian Articles Segregated by Olympic Relevance",
    labels={
        "Article Category": "Article Category",
        "Count": "Number of Articles"
    }
)

fig.update_traces(
    textposition="outside"
)

fig.update_layout(
    template="plotly_white",
    title_x=0.5,
    showlegend=False
)

fig.show()
```

Interpretation of Olympic Article Segregation Visualisation

The visualisation highlights the proportion of Olympic-related articles identified within the broader Guardian sports media dataset. Segregating Olympic-focused articles from general sports coverage helps reduce unrelated media noise and ensures that subsequent semi-structured analytics remain more relevant to the research objective.

A bar chart was selected for this visualisation because it provides a simple and interpretable comparison between Olympic-related and non-Olympic article volumes. This makes it easier to understand the relative concentration of Olympic-focused media coverage within the collected dataset.

From an analytical perspective, the segregation process strengthens the reliability of later techniques such as Count Vectorization, TF-IDF, and NMF topic modelling by focusing the analysis on articles more closely connected to Olympic participation, athletes, and Brisbane 2032 sporting narratives.

For ASC and Brisbane 2032 stakeholders, this filtered dataset provides a more targeted representation of Olympic-related media visibility and helps support deeper exploration of how media attention may relate to youth sporting engagement and long-term audience development.

10. Count Vectorization of Guardian Media Data

Count Vectorization is applied to the cleaned Guardian media headlines to identify the most frequently occurring words in the collected sports articles. This provides a simple frequency-based view of the dominant terms appearing across the media dataset.

This step supports the semi-structured analysis by showing which Olympic and sports-related themes appear most often. For ASC and Brisbane 2032 stakeholders, these keyword patterns provide an exploratory indication of media visibility around Olympic sports, athletes, and major sporting events.

```
# ================================================
# COUNT VECTORIZATION OF GUARDIAN MEDIA DATA
# ================================================

from sklearn.feature_extraction.text import CountVectorizer

count_vectorizer = CountVectorizer(
    stop_words="english",
    max_features=20
)

count_matrix = count_vectorizer.fit_transform(
    guardian_df["clean_headline"]
)

count_df = pd.DataFrame(
    count_matrix.toarray(),
    columns=count_vectorizer.get_feature_names_out()
)

word_counts = (
    count_df
    .sum()
    .sort_values(ascending=False)
    .reset_index()
)

word_counts.columns = ["keyword", "frequency"]

display(word_counts)
```

```
# ================================================
# VISUALISE COUNT VECTORIZATION RESULTS
# ================================================

plot_data = word_counts.sort_values(
    by="frequency",
    ascending=True
)

fig = px.bar(
    plot_data,
    x="frequency",
    y="keyword",
    orientation="h",
    text="frequency",
    title="Most Frequent Keywords in Guardian Sports Media Data",
    labels={
        "frequency": "Keyword Frequency",
        "keyword": "Keyword"
    },
    height=650
)

fig.update_traces(
    textposition="outside"
)

fig.update_layout(
    template="plotly_white",
    title_x=0.5,
    showlegend=False,
    margin=dict(l=120, r=80, t=80, b=60)
)

fig.show()
```

Interpretation of Count Vectorization Visualisation

The Count Vectorization visualisation highlights the most frequently occurring words within the Guardian sports media headlines related to Olympic and youth sporting discussions. Unlike TF-IDF, which measures contextual importance, Count Vectorization focuses purely on term frequency and provides a clearer overview of dominant media vocabulary across the dataset.

A horizontal bar chart was selected for this visualisation because it allows easier comparison of keyword frequencies while improving readability for longer sports-related terms. This format also helps present the relative prominence of different Olympic themes in a more interpretable manner for stakeholders.

From an analytical perspective, the visualisation helps identify which sporting concepts, athletes, and Olympic-related themes appear most consistently throughout the media dataset.

Frequently occurring terms may indicate areas receiving stronger media visibility, public attention, and broader sporting discussion.

For ASC and Brisbane 2032 stakeholders, these findings provide exploratory insight into the types of sports narratives currently dominating media coverage. Understanding these visibility patterns may help support future sport promotion strategies, youth engagement initiatives, and long-term Olympic audience development planning.

11. NMF Topic Modelling of Olympic Media Narratives

To further explore hidden thematic structures within the Olympic-related Guardian media dataset, Non-Negative Matrix Factorization (NMF) topic modelling is applied to the filtered article headlines.

Unlike Count Vectorization and TF-IDF, which focus on keyword frequency and importance, NMF helps identify groups of words that commonly appear together across multiple headlines. This allows broader sporting themes and Olympic-related narratives to emerge from the media dataset.

From a stakeholder perspective, topic modelling provides exploratory insight into the dominant sporting discussions currently receiving media attention. Understanding these recurring themes may help ASC and Brisbane 2032 stakeholders better understand how athlete narratives, Olympic visibility, and sporting events contribute to youth sporting interest and long-term audience engagement.

```
# ================================================
# NMF TOPIC MODELLING
# ================================================

from sklearn.decomposition import NMF
from sklearn.feature_extraction.text import TfidfVectorizer

# TF-IDF vectorization
tfidf_vectorizer = TfidfVectorizer(
    stop_words="english",
    max_features=100
)

tfidf_matrix = tfidf_vectorizer.fit_transform(
    olympic_articles_df["clean_headline"]
)

# Apply NMF
nmf_model = NMF(
    n_components=3,
    random_state=42
)
```

```
nmf_model.fit(tfidf_matrix)

# Extract feature names
feature_names = tfidf_vectorizer.get_feature_names_out()

# Display topics
topics = []

for topic_idx, topic in enumerate(nmf_model.components_):

    top_keywords = [
        feature_names[i]
        for i in topic.argsort()[:-11:-1]
    ]

    topics.append({
        "Topic": f"Topic {topic_idx + 1}",
        "Top Keywords": ", ".join(top_keywords)
    })

topics_df = pd.DataFrame(topics)

display(topics_df)

# ================================================
# VISUALISE NMF TOPIC STRENGTH
# ================================================

topic_strength = nmf_model.components_.sum(axis=1)

topic_strength_df = pd.DataFrame({
    "Topic": [
        "Topic 1",
        "Topic 2",
        "Topic 3"
    ],
    "Strength": topic_strength
})

fig = px.bar(
    topic_strength_df,
    x="Topic",
    y="Strength",
    text="Strength",
    title="Strength of NMF Topics within Olympic Media Coverage",
    labels={
        "Strength": "Topic Strength"
    }
)
```

```
fig.update_traces(
    textposition="outside"
)

fig.update_layout(
    template="plotly_white",
    title_x=0.5,
    showlegend=False
)

fig.show()
```

Interpretation of NMF Topic Modelling

The NMF topic modelling analysis identified several recurring thematic structures within the Olympic-related Guardian media dataset. Rather than focusing only on individual keyword frequency, the model grouped commonly associated words together to reveal broader sporting narratives appearing throughout the media coverage.

The identified topics suggest that Olympic media attention strongly centres around athlete performance, sporting events, international competition, and Olympic participation narratives. These recurring themes indicate that media visibility surrounding athletes and major sporting events remains highly prominent within contemporary sports reporting.

A bar chart was selected for the topic strength visualisation because it provides a simple comparison of the relative prominence of each discovered topic within the dataset. This allows clearer interpretation of which Olympic-related narratives appear more strongly throughout the media coverage.

From a stakeholder perspective, these findings provide exploratory insight into the dominant sporting narratives currently shaping Olympic-related media visibility. Understanding these thematic patterns may help ASC and Brisbane 2032 stakeholders support future youth engagement strategies, sport promotion initiatives, and long-term Olympic audience development planning.

12. Integrated Participation and Media Insights

To extend the investigation beyond participation statistics, media data from The Guardian API was analysed using TF-IDF (Term Frequency–Inverse Document Frequency). This technique helps identify the most important keywords appearing within sports-related media coverage associated with Olympic participation, youth sport, and Brisbane 2032.

While the earlier stages focused on structured AusPlay participation data, the Guardian dataset introduces an additional media perspective reflecting current sporting narratives and public attention. TF-IDF is valuable because it highlights words that are both frequent and contextually important within the article collection, rather than simply counting common terms.

Integrating media analytics with participation analysis provides a broader understanding of youth sporting engagement. It allows comparison between participation behaviour and media visibility while also identifying sports and Olympic themes receiving stronger public attention.

Overall, combining structured participation data with exploratory headline-based media analysis provides a more holistic understanding of youth engagement patterns and supports stronger strategic insight relevant to ASC planning and Brisbane 2032.

```
# ================================================
# CLEAN ARTICLE HEADLINES
# ================================================

guardian_df["clean_headline"] = (
    guardian_df["headline"]
    .astype(str)
    .str.lower()
    .apply(lambda x: re.sub(r"[^a-zA-Z\s]", "", x))
)

# Check columns and preview
print(guardian_df.columns)

display(
    guardian_df[
        ["headline", "clean_headline"]
    ].head(10)
)

# ================================================
# TF-IDF KEYWORD ANALYSIS
# ================================================

tfidf = TfidfVectorizer(
    stop_words="english",
    max_features=20
)

tfidf_matrix = tfidf.fit_transform(
    guardian_df["clean_headline"]
)

tfidf_scores = pd.DataFrame(
    tfidf_matrix.toarray(),
    columns=tfidf.get_feature_names_out()
)

keyword_scores = (
    tfidf_scores
    .mean()
    .sort_values(ascending=False)
```

```
    .reset_index()
)

keyword_scores.columns = ["keyword", "tfidf_score"]

display(keyword_scores)
```

Interpretation of TF-IDF Media Keyword Analysis

The TF-IDF keyword visualisation highlights the main themes currently dominating Olympic and sports-related media coverage connected to Brisbane 2032 and youth sport engagement. Unlike participation statistics alone, this analysis captures the topics receiving the strongest media visibility and public attention.

Keywords such as *Olympics*, *Olympic*, *athletics*, *athletes*, and *games* received the highest TF-IDF scores, indicating that Olympic-focused themes strongly influence contemporary sports reporting.

From an analytical perspective, this stage adds an external media dimension to the investigation. Media visibility may influence youth interest, sporting culture, and long-term audience engagement, making it useful to compare participation behaviour with broader media attention.

Overall, combining participation analysis with exploratory headline-based media analytics provides a more holistic understanding of youth sport engagement patterns and supports strategic insight relevant to ASC planning and Brisbane 2032.

```
# Keep top 15 keywords only
top_keywords = keyword_scores.head(15).sort_values(
    by="tfidf_score",
    ascending=True
)

fig = px.bar(
    top_keywords,
    x="tfidf_score",
    y="keyword",
    orientation="h",
    text="tfidf_score",
    title="Most Important Keywords in Guardian Sports Headlines",
    labels={
        "tfidf_score": "Average TF-IDF Score",
        "keyword": "Keyword"
    },
    height=600
)

fig.update_traces(
    texttemplate="%{text:.3f}",
    textposition="outside"
```

```
)

fig.update_layout(
    template="plotly_white",
    title_x=0.5,
    xaxis_title="Average TF-IDF Score",
    yaxis_title="Keyword",
    showlegend=False,
    margin=dict(l=120, r=80, t=80, b=60)
)

fig.show()
```

Interpretation of Olympic Media Themes

The TF-IDF analysis identified several dominant themes within Guardian media coverage related to Olympic-relevant sports. Frequently occurring terms such as *Olympics*, *athletes*, *games*, *championships*, and *gold* indicate that media attention strongly focuses on elite competition, athlete achievement, and international sporting success.

These themes are important because media visibility can influence public interest, youth motivation, and long-term audience engagement. High exposure to athletes and Olympic narratives may strengthen emotional connections between young audiences and sport participation.

Terms such as *Australian*, *athletics*, and *tennis* also suggest that national sporting identity and athlete representation remain highly visible within media coverage leading into Brisbane 2032.

Overall, the findings indicate that sports with strong youth participation are often associated with stronger media visibility. This relationship highlights how participation trends and exploratory headline-based media narratives may work together to support long-term Olympic engagement and ASC participation planning.

Integrating Participation and Media Insights

The AusPlay participation analysis and Guardian media analysis together provide a broader understanding of Olympic engagement patterns leading into Brisbane 2032.

The participation analysis identified sports such as swimming, basketball, gymnastics, tennis, and athletics as having strong youth engagement across multiple age groups. Similarly, the TF-IDF media analysis highlighted strong media attention surrounding Olympic athletes, competitions, and sporting success.

Combining both datasets helps demonstrate how youth participation and media visibility may work together to support long-term audience development and public sporting interest. Sports with both high participation and strong media presence may have greater potential for sustained Olympic engagement within Queensland communities.

Overall, integrating structured participation data with media-based thematic analysis provides a more strategic understanding of how grassroots involvement and public media narratives may influence future Olympic audience growth.

13. Ethical Considerations

While integrating participation data with media analysis provides broader insight into Olympic engagement patterns, several limitations should be considered. Media coverage may not always accurately represent actual participation behaviour, as certain sports can receive greater attention due to popularity, major events, or elite athlete success.

In addition, Guardian articles reflect editorial and media priorities, which may introduce bias toward highly visible sports while underrepresenting smaller or emerging activities. As a result, strong media attention should not automatically be interpreted as strong community participation.

The TF-IDF analysis also focuses only on keyword frequency and importance within headlines and does not fully capture deeper social context, sentiment, or audience interpretation. Therefore, the findings should be viewed as exploratory insights rather than definitive conclusions about youth sporting behaviour.

Overall, the analysis should be interpreted carefully and used to support inclusive and balanced sporting development strategies leading into Brisbane 2032.

14. AI Acknowledgement

Generative AI tools were used during the development of this investigation to support both technical and written components of the analysis. AI assistance was used to help generate and refine Python code related to data cleaning, visualisation, clustering analysis, Guardian API integration, and TF-IDF processing within the Jupyter Notebook environment.

AI tools were also used to assist with improving the clarity, structure, and professionalism of written narratives and analytical interpretations throughout the report. This included rephrasing explanations, improving vocabulary, refining markdown sections, and enhancing the overall readability of the analytical discussion.

All analytical decisions, interpretation of findings, selection of methods, and final review of the notebook were completed and verified by the author.

15. Strategic Recommendations

Based on the participation, clustering, and exploratory media visibility analyses, several strategic recommendations can be considered by the Australian Sports Commission (ASC) and Brisbane 2032 stakeholders.

Swimming, basketball, gymnastics, and athletics demonstrated consistently strong youth engagement across multiple demographic groups. Continued investment in grassroots development programs, school partnerships, and junior participation initiatives may therefore support long-term participation sustainability and Olympic audience development.

Lower-engagement sports such as archery, badminton, and table tennis may benefit from increased school-based exposure, community participation programs, and targeted promotional strategies designed to improve youth accessibility and participation awareness.

The media analysis also suggested that athlete-focused narratives and Olympic visibility remain highly influential within sports-related media coverage. Strategic use of athlete storytelling, digital campaigns, and Brisbane 2032 promotional initiatives may therefore strengthen youth engagement and long-term sporting interest.

Overall, combining structured participation analytics with exploratory media visibility analysis provides a broader evidence-based framework for supporting future sport investment, youth participation planning, and Olympic engagement strategies leading into Brisbane 2032.

16. Conclusion

This investigation combined structured AusPlay participation data with semi-structured Guardian API media data to explore youth engagement patterns across Olympic-relevant sports leading into Brisbane 2032.

The analysis identified sports such as swimming, basketball, gymnastics, athletics, and tennis as demonstrating strong youth participation across multiple age groups. Clustering and segmentation analysis further revealed clear differences in participation behaviour, highlighting variations in engagement intensity and long-term participation potential across sports.

The Guardian API and TF-IDF analysis extended the investigation beyond participation statistics by identifying dominant Olympic-related media themes and exploratory headline-based media visibility patterns. The findings suggested that sports receiving stronger media attention often align with activities showing higher youth engagement.

Overall, combining participation analytics with media analysis provided a broader understanding of how youth engagement and media visibility may collectively influence long-term sporting culture and Olympic audience development.

Based on these findings, ASC could prioritise high-engagement sports such as swimming and basketball within future youth participation initiatives while also supporting lower-engagement sports through targeted grassroots development and community promotion strategies leading into Brisbane 2032./